

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding (2)	Method Selection (2)	Solution Process (2.5)	Accuracy of Calculation (2)	Interpretation of Results (1.5)	Total(10)
Weightage	20%	20%	25%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _____Siva Charan (192525039)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Siva Charan

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO2 ASSESSMENT TOOL-1

1) Suppose that the data for analysis includes the age the age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

(a) What is the mean of the data? What is the median?

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$13 + 15 + 16 + 16 + 19 + 20 + 20 + 21 + 22 + 22 + 25 + 25 + 25 + 25 + 30 + 33 + 35 + 35 + 35 + 35 + 36 + 40 + 45 + 46 + 52 + 70$$

$$= \frac{809}{27} = 29.9629$$

$$\text{Median} = \frac{n+1}{2} = \frac{27+1}{2} = \frac{28}{2} = 14^{\text{th}} \text{ value}$$

$$14^{\text{th}} \text{ value} = 25 - \text{median value}$$

(b) What is the mode of the data? Comment on the data's modality?

25 & 35 are repeating for 4 times and it is binomial values.

(c) What is the midrange of the data?

$$\frac{\text{max} + \text{min}}{2} = \frac{13 + 70}{2} = \frac{83}{2} = 41.5$$

(d) Can you find the first quartile (q_1) and third (q_3) of the given data?

$Q_1 = \text{median value data}$

$$13 + 15 + 16 + 16 + 19 + 20 + 20 + 21 + 22 + 25 + 25 + 25 + 25$$

$$\frac{n+1}{2} = \frac{13+1}{2} = \frac{14}{2} = 7^{\text{th}} \text{ value.}$$

$$Q_3 = 30 + 33 + 33 + 35 + 35 + 35 + 35 + 36 + 40 + 45 + 45$$

$Q = 35$ the median of the upper 13 value is the 7th value.

$$Q_3 - Q_1 = 35 - 7 = 28$$

(e) Given the five number summary of the data?

min, Q_1 , median, Q_3 , max

13, 20, 25, 35, 70

$$13 + 15 + 16 + 19 + 20 + 21 + 22 + 25 + 30 + 33 + 35 + 36 + 40 + 45 + 45 + 50 + 52 + 70$$

2) Given the following measurements for the variable age 18, 22, 25, 42, 28, 43, 33, 35, 56, 28 standardize the variable by the following.

compute the mean absolute deviation of age.

$$= \frac{330}{10} = 33$$

$$|x - \bar{x}|$$

(m)	Age	$ n - x_i $
	18	15
	22	11
	25	8
	42	9
	28	5
	43	10
	33	6
	35	2
	56	23
	28	5

$$\sum |n - x_i| = 88/10 = 8.8$$

(3) Suppose that the data for analysis include the attribute age the age value for the data tuple are 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use smoothing by bin means to smooth the above data, using a bin depth of 3. illustrate your steps?

Ans: Total No. of Values = 27.

$$\text{Total No. of bins} = \frac{\text{total}}{\text{bin depth}} = \frac{27}{3} = 9$$

Bin	Value	Bin mean	Smoothed values
B1	13, 15, 16	44/3	14.67
B2	16, 19, 20	55/3	18.33
B3	20, 21, 22	63/3	21
B4	22, 25, 25	72/3	24
B5	25, 25, 30	80/3	26.67
B6	33, 33, 35	101/3	33.67
B7	35, 35, 35	105/3	35
B8	36, 40, 45	121/3	40.33
B9	46, 52, 70	168/3	56

(4) Use the two methods below to normalize the following group of data : 200, 300, 400, 600, 1000 (a) min-max normalization by setting min=0 and max=1 (b) z-score normalization

A) min-max Normalisation

$$x = \frac{x - \min}{\max - \min} = \frac{x - 200}{1000 - 200} = 800$$

Original Value	Calculation	Normalized
200	$200 - 200 / 800$	0
300	$300 - 200 / 800$	0.125
400	$400 - 200 / 800$	0.25
600	$600 - 200 / 800$	0.50
1000	$1000 - 200 / 800$	1.00

b) z-score Normalization.

$$z = \frac{x - \bar{x}}{\sigma}$$

$$\bar{x} = 200 + 300 + 400 + 600 + 1000 = \frac{2500}{5} = 500$$

Standard deviation.

x	$x - 500$	$(x - 500)^2$
200	-300	90,000
300	-200	40,000
400	-100	10,000
600	100	10,000
1000	500	2,50,000

$$\text{Total} = 400,000$$

$$\sigma = \sqrt{\frac{400,000}{5}} = \sqrt{80,000} = 282.8411$$